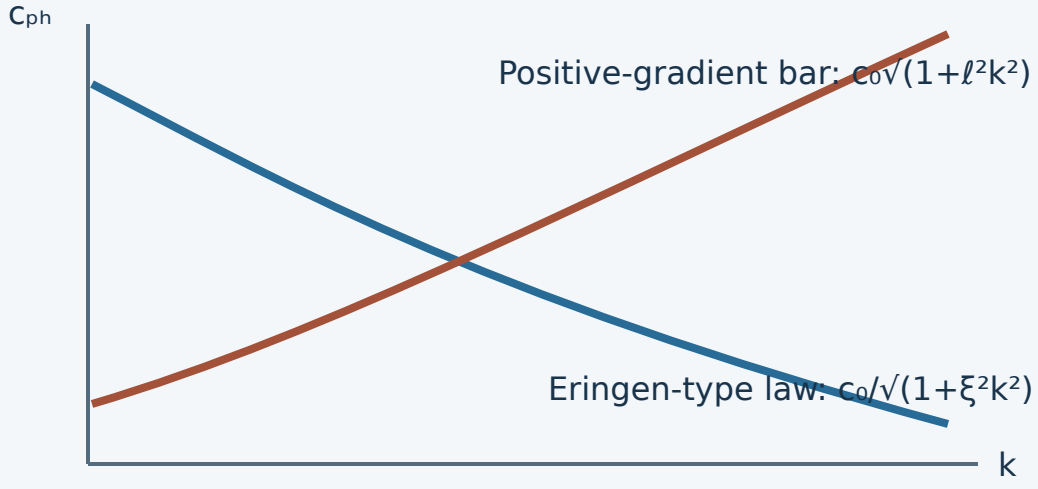
Spatial nonlocality and gradients: compare specified laws

One-dimensional phase speed versus wavenumber



Qualitative curves from the two displayed one-dimensional laws only.

Nonlocal law

$\omega = c_0k/\sqrt{(1+\xi^2k^2)}$

$\omega \to c_0/\xi$; c_{ph}, $c^g \to 0$

As $k \to \infty$ (for this law).

Positive gradient law

$c^2 = c_0^2(1+\ell^2k^2)$

Phase speed increases with k.

For the stated 1D bar model.

Energy flux and well-posedness are separate questions

Kernel-, domain-, and boundary-specific derivations are required; no universal extra flux is asserted.